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RESEARCH-ARTICLE

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Enhancing Microservices Anomaly Detection via Multimodal Data Fusion in the Wavelet Domain and Spatiotemporal Graph-based Diffusion Probabilistic Model

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Abstract

Microservices architecture has become increasingly popular in modern software systems, yet its complexity also poses challenges for failure diagnosis. Existing literature has deficiencies in multimodal data mining, mainly in two aspects: first, the mining of multimodal data is insufficient, failing to fully exploit the rich information contained in different types of data; second, the analysis of spatiotemporal features of multimodal data is not thorough, failing to fully explore the potential associations of data in the temporal and spatial dimensions. To address these issues, this paper proposes a new method for anomaly detection in microservices. We transform three types of modal data into the wavelet domain and achieve fusion among modalities based on an attention mechanism, fully utilizing the inherent characteristics of multimodal data in the frequency and temporal domains. This fusion process can extract new information that does not exist in the original modalities, thereby enhancing the model's ability to detect anomalies. In addition, we propose a new diffusion probabilistic model (DDPM) based on spatiotemporal graphs, which combines spatiotemporal learning capabilities with the uncertainty measurement of DDPM to generate future samples in a non-autoregressive manner, achieving multi-horizon prediction. Experimental results show that our method significantly improves anomaly detection performance on three public datasets and a real-world production system, demonstrating its effectiveness in microservices anomaly detection.

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CCS Concepts

• **Computing methodologies** → **Anomaly detection**; • **Software and its engineering** → *Software performance*.

Keywords

Microservices Anomaly Detection; Multimodal Data Fusion; Wavelet Domain Attention Mechanism

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1 Introduction

Microservice architecture communicates through lightweight mechanisms and has gained significant popularity in recent years [2, 7]. However, the large scale of microservice systems presents challenges, particularly in anomaly detection. Anomalies in these systems can lead to data loss, business disruptions, financial losses, and reduced operational efficiency [1]. Recognizing the importance of service reliability, major cloud providers such as Amazon Web Services (AWS) and Microsoft Azure have made significant investments in anomaly detection for microservices [20, 31].

To ensure effective anomaly detection, current research predominantly relies on analyzing three critical data modalities: metrics (time-series data), logs (textual structured data), and traces (graph-structured data). Most existing anomaly detection methods are based on single-modality data [28]. However, previous studies [15, 19, 26] have highlighted that these methods are unable to capture anomaly information related to certain faults. To address the limitations of single-modal data, methods [11, 22, 32] use multimodal data to achieve more effective failure diagnosis. They first preprocess heterogeneous data to extract a more reasonable representation of multimodal features [11, 15, 22, 26, 32]. These unified features are then used as inputs for deep learning or other machine learning models to detect anomalies. Multimodal data

integrates information from various sources, offering a more comprehensive reflection of the system's operational state; however, several limitations persist in existing methods:

- **Limitation 1: Insufficient exploitation on multi-modal data.** Current multimodal anomaly detection approaches inadequately explore cross-modal relationships. Most approaches [22, 32] employ early fusion by simply concatenating multimodal data as multiple channels. However, they fail to capture complex intermodal relationships in sophisticated multimodal datasets. While later fusion methods [15, 19, 25, 26] in Table 1 primarily focus on attention-based fusion of metrics and logs, they generally neglect three-modal fusion scenarios. More critically, existing research overlooks frequency-domain analysis of multimodal data. Temporal analysis focuses on direct observations at specific time points, potentially missing periodic patterns that frequency analysis could reveal. This limitation is particularly problematic for non-stationary microservice data, where temporal-domain analysis alone may be insufficient to capture the full complexity of the data.
- **Limitation 2: Lack of spatiotemporal feature learning.** Accurately identifying anomalous patterns requires modeling both spatial relationships (e.g., service invocation dependencies) and temporal dynamics (e.g., variations over time). However, existing approaches exhibit critical gaps in this regard. For instance, Deeptralog [30] fails to incorporate temporal features, disregarding essential temporal variations in traces. Similarly, studies such as [15, 19, 32] neglect spatial dependencies across modalities in Table 1. Current spatiotemporal modeling [11, 22, 25, 26] predominantly relies on deterministic approaches, which compute future states without accounting for randomness. Microservice architectures create data uncertainty from diverse sources and complex dependencies. Spatiotemporal models for multimodal microservice data should incorporate uncertainty by using probability distributions instead of deterministic outputs.

To systematically address the two identified limitations, we propose **FUSION**, a novel framework that integrates **F**requency-domain multimodal fusion based on attention mechanisms with **U**ncertainty-aware **S**patiotemporal **I**ntegration for **O**nline anomaly detectio**N**. This paper is the first to address the challenges of multimodal data fusion and anomaly detection on spatiotemporal graphs. **Limitation 1** is approached through FUSION's feature fusion in the wavelet domain, which effectively captures both frequency-domain and time-domain features of multimodal data. More specifically, we decompose multimodal data into different frequency bands (wavelets) with different resolutions represented as wavelet coefficients to enable more sophisticated time series feature extraction. Then, we apply a correlation-based attention mechanism by calculating Pearson Correlation Coefficients among variables across modalities. The mechanism adjusts the weights of their wavelet coefficients for highly correlated variables, emphasizing the most useful information from the wavelet domain. Finally, we reconstruct the weighted coefficients into a frequency-enhanced feature representation. For **Limitation 2**, we employ a novel spatiotemporal graph-based Denoising Diffusion Probabilistic Model (DDPM), which combines spatiotemporal learning with uncertainty measurements. FUSION leverages a U-structure-based architecture [23] to capture temporal dependencies and spatial correlations. Unlike traditional DDPMs, FUSION employs randomized diffusion steps to more effectively capture spatiotemporal dynamics and improve denoising during the reverse process. Furthermore, FUSION generates future samples in a non-autoregressive manner, enabling efficient multi-horizon predictions all at once. This model offers both historical data reconstruction and future prediction, providing key benefits for fault diagnosis. Reconstructing past data helps analyze failure trends and supports problem-solving. Its predictive abilities provide early fault warnings, enabling preventive maintenance that reduces risks and improves system stability.

Table 1: Compared to FUSION, existing methods did not comprehensively address multimodal anomaly detection (Temp for Temporal, Spat for Spatio, and Freq for Frequency).

Approach	Modality			Feature Extraction				
	Metric	Log	Trace	Modal fusion	Modal attention	Temp	Spat	Freq
DeepTraLog	✗	✓	✓	✗	✗	✗	✓	✗
Hades	✓	✓	✗	Later fusion	cross-attention	✓	✗	✗
MADMM	✓	✓	✗	Later fusion	cross-attention	✓	✓	✗
MISP	✓	✓	✗	Later fusion	cross-attention	✓	✗	✗
Medicine	✓	✓	✓	Later fusion	channel attention	✓	✓	✗
AnoFusion	✓	✓	✓	Early Fusion	✗	✓	✗	✗
Triple	✓	✓	✓	Early Fusion	✗	✓	✓	✗
MSTGAD	✓	✓	✓	✗	✗	✓	✓	✗
FUSION	✓	✓	✓	**Later fusion**	**wavelet fusion**	✓	✓	✓

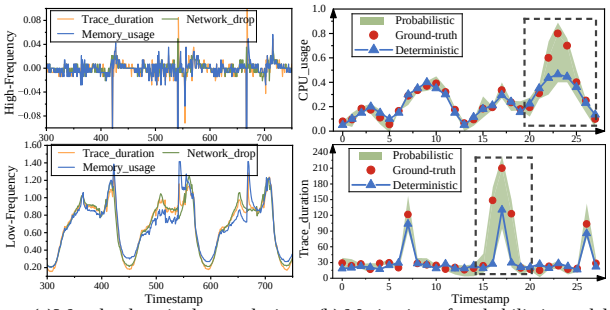
We conducted extensive studies to evaluate FUSION on three open-source platforms: AIOPS2022 [4], GAIA [5], and SocialNetwork [9], and a real-world production microservice system. Benefiting from the use of multimodal data, FUSION achieves a notable improvement, increasing F1-score by 2.70%–129.27%, 3.62%–137.37%, 2.70%–86.16%, and 10.21%–168.83% on four datasets, respectively. In summary, this paper has made the following primary contributions.

- We are the first to propose feature fusion for multimodal data in the wavelet domain, leveraging a Pearson correlation-based attention mechanism to enhance anomaly detection by combining both temporal and frequency features.
- We tackle the problem of probabilistic spatiotemporal anomaly detection from a conditional diffusion perspective with the first shot. FUSION can effectively model the complex spatiotemporal dependencies and intrinsic uncertainties within multimodal data.
- We conduct extensive experiments to demonstrate that FUSION is more effective than state-of-the-art anomaly detectors. We have made the implementation code publicly available at https://github.com/AmadeusClo/FUSION.

2 Motivation

2.1 Why Fusion in Wavelet Domain?

Discrete Wavelet Transform (DWT) effectively decomposes non-stationary multimodal data into wavelet coefficients across frequency bands, capturing both short-term fluctuations (detail coefficients) and long-term trends (approximation coefficients). Our analysis reveals a significant correlation between high-frequency and low-frequency components across multimodal data, which exhibit consistent periodicity and synchronized peak-valley variations in Figure 1(a). The wavelet domain feature fusion offers



(a)Wavelet domain data analysis (b) Motivation of probabilistic model

Figure 1: Multimodal data analysis and probabilistic prediction. (a) The wavelet decomposition of multimodal data emphasizes the correlation of wavelet domain.(b) Comparing deterministic methods (e.g., Spatiotemporal convolution) with probabilistic approaches (e.g., DDPM) and demonstrating the superiority of probabilistic methods particularly when CPU or trace latency spikes occur in the black dotted box area.

two key advantages: First, it preserves the unique characteristics of each modality while leveraging their complementary relationships. Second, it enhances the system's resilience to noise, as the presence of multiple modalities ensures that valuable insights can still be extracted even when one modality is compromised. This dual advantage improves the model's robustness, particularly in complex and dynamic signal environments where data quality and consistency may vary.

2.2 Why DDPM?

Recent multimodal anomaly detection techniques are mainly deterministic, accurately predicting future data [11, 22, 26, 32]. However, these methods face challenges with spatiotemporal uncertainties, affecting their real-world decision-making effectiveness. We adopt probabilistic models over deterministic ones for the uncertainty modeling ability. Microservice multimodal data inherently involves uncertainty due to diverse data sources, complex dependencies, varying environments, and data noise or incompleteness. Probabilistic models address this by representing data states through probability distributions rather than a single deterministic value, naturally capturing the uncertainty as shown in Figure 1(b).

3 Proposed Approach

Figure 2 presents the overall architecture of FUSION, which consists of ***multimodal data serialization***, ***multimodal wavelet fusion*** and ***conditional diffusion network***. First, preprocess the multimodal data. Next, these three modalities are combined in the wavelet domain based on an attention-based mechanism. The third stage is the integration of a U-structure-based denoising process during the diffusion phase. This enhances the precision of the denoising and effectively captures the spatiotemporal dependencies within the multimodal data. Next, we elaborate on these stages.

3.1 Multimodal Data Serialization

3.1.1 Metric Serialization. Microservices have a set of metrics representing a multivariate time series. In this paper, we focus on multivariate time-series data, denoted as $\hat{H}_i^M = \{h_{i,0}^M, \ldots, h_{i,T}^M\} \in \mathbb{R}^{F_M \times T}$, representing a sequence of timestamped observations of size T for the i-th instance and F_M is the number of metrics.

3.1.2 Log Serialization. FUSION uses the Drain3 [25] algorithm, which excels in performance and efficiency for log template extraction. The serialization process consists of two main steps:

(1) Log Template Clustering. To manage diverse log variations, we cluster log templates to group similar entries and reduce redundancy. When software updates, we classify new log templates by comparing them to existing cluster centroids to determine their membership or the need for new clusters. We utilize the bert-base-uncased [8] model for embedding vectors and applying the DBSCAN [24] algorithm, resulting in $F_L - 1$ clusters.

(2) Log Entry Serialization. The system calculates the distance between the sentence embedding vectors of log entries and cluster centroids to determine their categories. If a log entry doesn't match a cluster, it's identified as an anomalous template and assigned to a separate category, resulting in F_L total categories. To align this analysis with system metrics, we count the occurrences of each log category over time, generating a log representation $\hat{H}_i^L = \{h_{i,0}^L, \ldots, h_{i,T}^L\} \in \mathbb{R}^{F_L \times T}$.

3.1.3 Trace Serialization. In a microservice system, a trace captures user request executions as spans, which include attributes like trace ID, request type, span ID, parent span ID, service instance ID, start time, and duration. We analyze these spans sequentially to extract structural information and dependencies. Duration is then transformed into a multivariate time series by calculating the average latency, total requests, and status code frequencies per minute. The resulting trace multivariate time series is represented as $\hat{H}_i^{Tr} = \{h_{i,0}^{Tr}, \ldots, h_{i,T}^{Tr}\} \in \mathbb{R}^{F_{tr} \times T}$, where F_{tr} is the number of spans for instance i.

3.1.4 Multimodal Data Construction. We perform z-score standardization [26] using a sliding historical window on a time series modal. For each instance i at time t, we concatenate the features into a fused feature vector $H_i^{modal} = \left(\hat{H}_i^M \| \hat{H}_i^L \| \hat{H}_i^{Tr}\right) = \{x_{i,0}^{modal}, \ldots, x_{i,T}^{modal}\} \in \mathbb{R}^{F_{modal} \times T}$, where $F_{modal} = F_M + F_L + F_{Tr}$. This achieves serialization across metrics, logs, and traces.

3.2 Multimodal Wavelet Fusion

3.2.1 Discrete Wavelet Transform. The DWT module transforms feature vectors H_i^{modal} into multi-scale frequency representations by breaking down input signals into wavelets. This process captures the original data's frequency and time characteristics. As depicted in Figure 2(b), DWT can be performed multiple times, and each DWT uses a high-pass filter $\mathbf{f}$ and a low-pass filter $\mathbf{g}$ to decompose a time series signal H_i^{modal} into different resolutions. The outputs of the high-pass and low-pass filters at layer l are denoted as cD_l and cA_l, respectively. l indicates the l-th decomposition and $cA_0 = H_i^{modal}$.

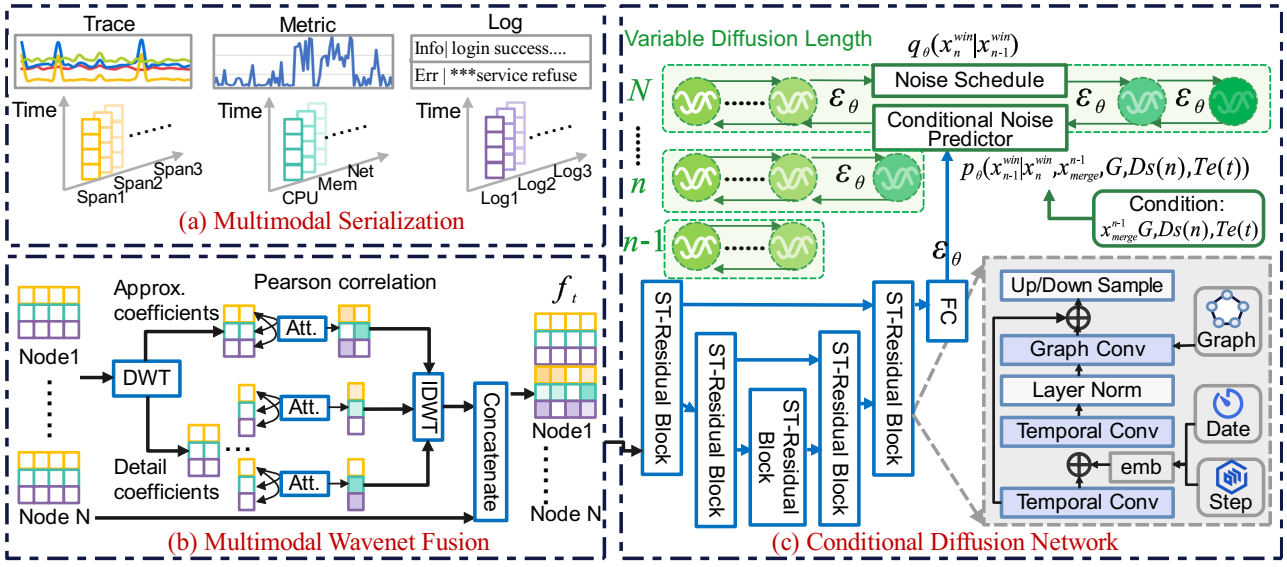


Figure 2: The overview training and diagnosis framework of *FUSION*. (a) multimodal data serialization: prepare for the future graph structure's construction into time series; (b) multimodal wavelet fusion: decompose multimodal data into wavelet coefficients, which are then fused via Pearson Correlation-Based Attention to identify significant features within and across the modalities. (c) conditional diffusion model: model uncertainties with multi-scale temporal dependencies and spatial correlations enabling efficient non-autoregressive prediction.

Specifically, we have

$$\begin{aligned}\mathbf{cD}_l = \mathbf{f} * \mathbf{cA}_{l-1} &= \sum_{m=1}^{\frac{T_w}{2^{l-1}}} \mathbf{f}[2s-m]\mathbf{cA}_{l-1}[m], \quad s = 1, \cdots, \frac{T_w}{2^l} \\ \mathbf{cA}_l = \mathbf{g} * \mathbf{cA}_{l-1} &= \sum_{m=1}^{\frac{T_w}{2^{l-1}}} \mathbf{g}[2s-m]\mathbf{cA}_{l-1}[m], \quad s = 1, \cdots, \frac{T_w}{2^l}\end{aligned} \tag{1}$$

where T_w represents the length of H_i^{modal} in a window size, the length change to $\frac{T_w}{2^l}$ after decomposing $l-1$ times and s represents the scale. The choice of filters $\mathbf{f}$ and $\mathbf{g}$ is based on the wavelet basis; once it's defined, the forms of $\mathbf{f}$ and $\mathbf{g}$ are also determined. The detail coefficients $\mathbf{cD}$ capture short-term trends and nuances, while the approximate coefficients $\mathbf{cA}$ indicate long-term trends. Moreover, deeper decomposition enhances the frequency resolution.

Let $C = \{\mathbf{C}_1, \mathbf{C}_2, \ldots, \mathbf{C}_l, \mathbf{C}_{l+1}\}$ represent the layered wavelet coefficients where each layer contains each variable's corresponding coefficients, denoted as follows:

$$\begin{aligned}\mathbf{cD}_l &= [\mathbf{cD}_l^{(1)}; \ldots; \mathbf{cD}_l^{(F_{modal})}] \in \mathbb{R}^{F_{modal} \times \frac{T_w}{2^l}}, \quad l \in [1, L] \\ \mathbf{cA}_L &= [\mathbf{cA}_L^{(1)}; \mathbf{cA}_L^{(2)}; \ldots; \mathbf{cA}_L^{(F_{modal})}] \in \mathbb{R}^{F_{modal} \times \frac{T_w}{2^l}}\end{aligned} \tag{2}$$

3.2.2 Fusion with Pearson Correlation-Based Attention. We introduce an attention mechanism based on the Pearson correlation coefficient to capture the correlation within and across the modalities. First, we compute the Pearson correlation coefficient matrix between the variables $R \in \mathbb{R}^{F_{modal} \times F_{modal}}$, of which F_{modal} is the number of variables. The Pearson correlation coefficient measures the linear correlation between variables and is defined as follows:

$R_{F_i,F_j} = \text{cov}(\mathbf{x}_i^{F_i}, \mathbf{x}_i^{F_j})/(\sigma_{\mathbf{x}_i^{F_i}} \sigma_{\mathbf{x}_i^{F_j}})$, where $\text{cov}(\mathbf{x}_i^{F_i}, \mathbf{x}_i^{F_j})$ denotes the covariance between variables F_i and F_j of feature matrix $\mathbf{H}_i^{modal}$ for instance i, σ is their standard deviations operation.

Based on the correlation matrix R_{F_i,F_j}, This mechanism learns the attention weights for each wavelet coefficient between variables through training, dynamically integrating the wavelet coefficients of other variables with itself to compute a vector that incorporates information from other variables. The attention weights are learned through model training, enabling the model to more flexibly capture complex dependencies between variables. Specifically, for the k-th wavelet coefficient $\mathbf{cD}_l^{(k)}$ at the l-th layer, we learn the attention weights $\alpha_{F_i,F_j,l}$ between it and the wavelet coefficients of other variables through training.

$$\alpha_{F_i,F_j,l} = \frac{\exp\left(\text{Att}(\mathbf{cD}_l^{(F_i)}, \mathbf{cD}_l^{(F_j)})\right)}{\sum_{m=1}^{F'_{modal}} \exp\left(\text{Att}(\mathbf{cD}_l^{(F_i)}, \mathbf{cD}_l^{(m)})\right)}, \tag{3}$$

where $\text{Att}(\mathbf{cD}_l^{(F_i)}, \mathbf{cD}_l^{(F_j)})$ is a learnable weight function that measures the strength of the correlation between variable F_i and F_j. The attention weight $\alpha_{F_i,F_j,l}$ reflects the importance of variable F_i to variable F_j.

Next, we use a weighted sum to integrate the wavelet coefficients of other variables with itself, computing the context vector $\boldsymbol{vD}_l^{(F_i)}$ for variable k:

$$\boldsymbol{vD}_l^{(F_i)} = \mathbf{cD}_l^{(F_i)} + \sum_{m=1}^{F'_{modal}} \alpha_{F_i,F_j,l} \cdot \mathbf{cD}_l^{(m)}. \tag{4}$$

Here, $\mathrm{cD}_l^{(F_i)}$ is the coefficient of variable F_i itself, while $\sum_{m=1}^{F'_{modal}} \alpha_{F_i,F_j,l} \cdot \mathrm{cD}_l^{(m)}$ represents the contribution of variables m to variable F_i.

Finally, we concatenate the context vectors from all layers to obtain the feature vector representing the multi-scale frequency:

$$\begin{aligned} \mathbf{vD}_l &= [\mathbf{vD}_l^{(1)}; \ldots; \mathbf{vD}_l^{(F_{modal})}] \in \mathbb{R}^{F_{modal} \times \frac{T_w}{2^l}}, \quad l \in [1, L] \\ \mathbf{vA}_L &= [\mathbf{vA}_L^{(1)}; \ldots; \mathbf{vA}_L^{(F_{modal})}] \in \mathbb{R}^{F_{modal} \times \frac{T_w}{2^l}} \end{aligned} \tag{5}$$

we apply the Inverse Discrete Wavelet Transform (IDWT) to reconstruct their corresponding sequence in the time domain. The process can be formulated as follows:

$$\mathrm{vA}_{l-1} = IDWT(\mathbf{vD}_l, \mathbf{vA}_l) = \mathbf{f}' \star \mathbf{vD}_l + \mathbf{g}' \star \mathbf{vA}_l \tag{6}$$

where $\mathbf{f}'$ and $\mathbf{g}'$ are the synthesis versions of $\mathbf{f}$ and $\mathbf{g}$. Then, $H_i^{merge} = \mathbf{vA}_0$ is the reconstructed time series of the i-th instance.

3.3 Conditional Diffusion Network

3.3.1 Framework of Conditional Diffusion Model. After obtaining H_i^{merge} and H_i^{modal}, we can integrate these three modalities together by constructing a microservice graph to virtually represent the status and scheduling relationships among service instances of a real-world microservice system. In this graph, nodes represent service instances, and edges represent scheduling relationships between different instances. Let $\mathcal{G} = \{\mathcal{V}, \mathbf{A}\}$ represent a graph with V nodes, where $\mathcal{V}$ is the node set and $\mathbf{A} \in \mathbb{R}^{V \times V}$ is a weighted adjacency matrix to describe the graph topology. Let $\mathbf{x}_t \in \mathbb{R}^{F_{modal} \times V}$ denote F_{modal}-dimensional signals generated by the i-th nodes at time t. Given historical graph signals $\mathbf{x}^h = [\mathbf{x}_1, \cdots, \mathbf{x}_{T_h}]$ of T_h time steps and the graph $\mathcal{G}$ as inputs, we aims at learning a function $\mathcal{F}$ to predict future graph signals $\mathbf{x}^p$, which is formulated as:

$$\mathcal{F} : (\mathbf{x}^h; \mathcal{G}) \rightarrow [\mathbf{x}_{T_h+1}, \cdots, \mathbf{x}_{T_h+T_p}] := \mathbf{x}^p, \tag{7}$$

where T_p is the forecasting horizon. In this study, we focus on the task of probabilistic spatiotemporal graph (STG) forecasting, which aims to estimate the distribution of future graph signals.

The original DDPM is designed to generate images from white noise without any conditions. However, this approach does not align with our task, where future signals are generated based on their historical data. To address this for STG forecasting (STG), we extend the DDPM to create a conditional version by making several modifications to the reverse process.

Training. Given the multimodal fusion data H_i^{merge}, construct the observed values $\mathbf{x}_{\mathbf{merge}}^{\mathbf{win}} = [\mathbf{x}_1, \cdots, \mathbf{x}_{T_h+T_p}]$ in time window $T_h + T_p$ for node i. Thus, the condition is $x_{\mathrm{merge}}^{\mathrm{win}}$, graph $\mathcal{G}$, and the target x_0^{win}, we sample noise targets $x_n^{\mathrm{win}} = \sqrt{\alpha_n} x_0^{\mathrm{win}} + \sqrt{1-\alpha_n}\epsilon$, and then train ϵ_θ using the loss function defined in Equation **??**. The loss function integrates the reconstruction of historical data with future estimations, maximizing the use of historical information for effective data distribution modeling. Additionally, it provides a unified framework for a range of spatiotemporal graph (STG) tasks, such as prediction, generation, and interpolation [21].

Inference. The inference process employs the trained denoising function ϵ_θ to sample x_{n-1}^{win} step by step according to Equation (13), under the guidance of $x_{\mathrm{merge}}^{\mathrm{win}}$ and $\mathcal{G}$. It is important to note that, unlike previous diffusion-based models like TimeGrad [21], which require executing the reverse diffusion process T_p times to predict the future T_p steps, FUSION only requires a single reverse diffusion process to make its predictions. This approach significantly accelerates the inference speed.

3.3.2 Conditional Generation with Unet-based Denoising Network. FUSION employs an Unet-based architecture in the temporal dimension to capture temporal dependencies at various granularities and utilizes GNN to model the spatial correlations. In Figure 2(c), we illustrate our denoising network that processes five dynamic inputs: x_{merge}^{all}, x_n^{win}, n, $\mathcal{G}$, and T, producing a clear output ϵ. The process begins by combining $x_n^{win} \in \mathbb{R}^{F_{modal} \times V \times T}$ with $x_{merge}^{win} \in \mathbb{R}^{F_{modal} \times V \times T}$ along the temporal dimension, resulting in a new tensor, $\widetilde{x}_n^{win} \in \mathbb{R}^{F_{modal} \times V \times 2T}$. This tensor is then projected to a higher-dimensional representation $X \in \mathbb{R}^{C \times V \times 2T}$ using a linear layer, where C is the projected dimension. X is subsequently processed through Spatio-temporal Residual Blocks (ST-Residual Blocks), designed to capture temporal and spatial dependencies. The input to the k-th block is denoted as $X_k \in \mathbb{R}^{C \times V \times T}$, with $X_0 = X$, reflecting the time dimension length T.

Temporal Dependency Modeling. In Figure 2(c), each ST-Residual Block processes the input X_k through a TCN [22] to capture temporal dependencies. This TCN uses a one-dimensional gated causal convolution with kernel size K and padding to ensure the output matches the input dimensions. The convolution kernel $\Gamma_T \in \mathbb{R}^{K \times C_{\mathrm{in}}^t \times C_{\mathrm{out}}^t}$ maps X_k to outputs P_i and Q_i in $\mathbb{R}^{C_{\mathrm{out}}^t \times V \times T}$. The temporal gated convolution can be defined as:

$$\Gamma_T(X_k) = P_i \odot \sigma(Q_i) \in \mathbb{R}^{C_{\mathrm{out}}^t \times V \times T}, \tag{8}$$

$\odot$ represents the Hadamard product, and σ is the sigmoid activation function. The term $\sigma(Q_i)$ is a gate that filters relevant information from P_i to the next layer. The output of the TCN is denoted as $\overline{X}_k$.

Spatial Dependency Modeling. Graph Convolutional Networks (GCNs) are used to extract meaningful features in the spatial domain [22]. The graph convolution is defined as:

$$\Gamma_G(\overline{X}_k) = \sigma\left(\Phi\left(A_{\mathrm{gcn}}, \overline{X}_k\right)\mathbf{W}_i\right), \tag{9}$$

where $\mathbf{W}_i \in \mathbb{R}^{C_{\mathrm{in}}^{\mathcal{G}} \times C_{\mathrm{out}}^{\mathcal{G}}}$ is a trainable parameter, and σ is an activation function. The aggregation function $\Phi(\cdot)$ combines features from neighboring nodes into the target node. We use the vanilla GCN [27], which defines a symmetric normalized summation function as: $\Phi_{\mathrm{gcn}}\left(A_{\mathrm{gcn}}, \overline{X}_k\right) = A_{\mathrm{gcn}}\overline{X}_k$, where $A_{\mathrm{gcn}} = D^{-\frac{1}{2}}(A+I)D^{-\frac{1}{2}} \in \mathbb{R}^{V \times V}$ is the normalized adjacency matrix of graph $\mathcal{G}$. Here, I is the identity matrix and D is the diagonal degree matrix with $D_{ii} = \sum_j (A+I)_{ij}$. We also reshape the TCN layer output into $\overline{X}_k \in \mathbb{R}^{V \times C_{\mathrm{in}}^{\mathcal{G}}}$, where $C_{\mathrm{in}}^{\mathcal{G}} = T_i \times C_{\mathrm{out}}^t$, and feed this feature into the GCN.

Diffusion Steps Embedding. FUSION utilizes the randomness of diffusion steps, which enhances the model's generative abilities, diversity, and robustness, especially with complex and noisy datasets. To optimize performance, we incorporate diffusion steps as auxiliary information, using positional encoding to represent the noise level $n \in [1, N]$ and process it through transformer positional

embedding.

$$Ds(n) = \left[\ldots, \cos\left(\frac{n}{r^{-\frac{2d}{D}}}\right), \sin\left(\frac{n}{r^{-\frac{2d}{D}}}\right), \ldots\right]^T, \tag{10}$$

where $d = 1, \cdots, D/2$ is the dimension of the embedding (typically set to 32), and r is a large constant (e.g., 10000). This embedding method captures the stochasticity of the diffusion steps and incorporates it as a conditional input to the model, thereby enhancing its generative capability and diversity.

Relative Time Embedding. We introduce a relative time embedding technique to address irregular time intervals. This technique captures the relative temporal relationships between time series elements through multi-resolution embeddings (days, hours, minutes). Specifically, the comprehensive embedding of a time series element is represented as:

$$Te(t) = E_D(t^d) + E_H(t^h) + E_M(t^m) \tag{11}$$

where days t^d, hours t^h, and minutes t^m can be calculated from t, respectively. $E_D = \left[\sin\left(\frac{2\pi t^d}{365}\right), \cos\left(\frac{2\pi t^d}{365}\right)\right]$, $E_H = \left[\sin\left(\frac{2\pi t^h}{365}\right), \cos\left(\frac{2\pi t^h}{365}\right)\right]$ and $E_M = \left[\sin\left(\frac{2\pi t^m}{365}\right), \cos\left(\frac{2\pi t^m}{365}\right)\right]$ represent the embedding functions for days, hours, and minutes. This embedding approach captures the periodic nature of time while preserving its relative aspects, allowing the model to better understand complex temporal relationships in time series data and enhancing its analytical performance.

3.4 Anomaly Evaluation

In practical fault prediction, to obtain the reconstructed historical values and future predictions, we sample the P_s data from the probability distribution in the reverse prediction process of the diffusion model over P_s/α steps. We employ a sampling acceleration strategy to enhance this process's efficiency.

Using a large number of diffusion steps N (e.g., $N = 1000$ [10]) helps the forward process approximate a Gaussian distribution, which is useful for the reverse denoising process. However, a high N can lead to inefficient sampling due to the need to execute all iterations sequentially. To improve efficiency, we adopt the selective sampling strategy from [33], which samples a subset $\{\tau_1, \cdots, \tau_M\}$ of M diffusion steps. The accelerated sampling process is defined as follows:

$$\begin{aligned} x_{\tau_{m-1}} = \sqrt{\alpha_{\tau_{m-1}}}\left(\frac{x_{\tau_m} - \sqrt{1-\alpha_{\tau_m}}\epsilon_\theta^{(\tau_m)}}{\sqrt{\alpha_{\tau_m}}}\right) \\ + \sqrt{1-\alpha_{\tau_{m-1}}-\sigma_{\tau_m}^2}\cdot\epsilon_\theta^{(\tau_m)} + \sigma_{\tau_m}\epsilon_{\tau_m} \end{aligned} \tag{12}$$

$\epsilon_{\tau_m} \sim \mathcal{N}(0, I)$ denotes independent standard Gaussian noise related to x_n, while the parameter σ_{τ_m} governs the stochastic nature of the denoising process. We define $\sigma_n = \sqrt{\frac{1-\alpha_{n-1}}{1-\alpha_n}}\sqrt{1-\frac{\alpha_n}{\alpha_{n-1}}}$ for all diffusion steps to comply with the Denoising Diffusion Probabilistic Model (DDPM). When the sampling trajectory length is much shorter than N, computational efficiency increases significantly. Moreover, the data from the last α reverse steps, x_i^{win} (with $i \in \{1, \ldots, P_s/\alpha\}$), provides a good approximation of the target, improving overall approximation quality. Thus, to estimate the data distribution from the trained model, i.e., getting P_s samples, we can reduce the number of reverse diffusion processes from P_s to P_s/α.

Table 2: Basic information about three datasets.

Data	Failure type	Cases	Percent	Data	Failure type	Cases	Percent
AIOPS 2022	File system read	21	12.07%	SN	Network delay	42	29.37%
	File system write	14	8.05%		Network loss	47	32.87%
	Network delay	19	10.92%		CPU stress	54	37.76%
	Network corruption	20	11.49%	MMS	CPU overload	26	9.81%
	Network duplicate	23	13.22%		Memory stress	19	7.17%
	Network drop packet	26	14.94%		High disk	24	9.06%
	CPU load	20	11.49%		Disk partition full	36	13.58%
	Memory load	19	10.92%		I/O latency	23	8.68%
	Container pause	12	6.90%		Database failure	14	5.28%
GAIA	login failure	527	47.95%		Resource exhaustion	38	14.34%
	Memory anomalies	505	45.95%		High packet loss	22	8.30%
	File moving program	36	3.28%		Network partitioning	18	6.79%
	Memory freed label	16	1.46%		Hardware Failures	32	12.08%
	Access denied	15	1.36%		Configuration Errors	13	4.91%

We utilize the Continuous Ranked Probability Score (CRPS) [13] as the abnormal score for each instance during the testing process. CRPS quantifies the degree of compatibility between a cumulative distribution function (CDF) F and an observation x for each modal data serialization and is calculated as $(F, x) = \int_{\mathbb{R}}(F(z) - \mathbb{I}\{x \leq z\})^2 \mathrm{d}z$. The abnormal score of i-instance can be calculated by summing the (F, x) of three modalities [6], resulting in $\hat{F}_{\text{sum}}(t)$ for each time point. The findings are subsequently calculated as the average over the prediction horizon. $AbScore = \mathbb{E}_t\left[\text{CRPS}\left(\hat{F}_{\text{sum}}(t), \sum_{i=1}^{D} x_{i,t}\right)\right]$.

Anomalies are observations that deviate from expected patterns in historical data. They occur when the expected conditional distribution probability of a prediction length T_p significantly differs from the actual value. FUSION calculates the anomaly severity using the AbScore. To trigger an anomaly event, it's crucial to select a threshold. In this study, we use a non-parametric, dynamic unsupervised threshold selection method (NDT) as introduced in [12]. NDT first creates the set e given a set of abnormal scores s of historical observations, where z is an ordered set of positive values.

$$\mathbf{e} = \mu(s) + \mathbf{z}\sigma(s). \tag{13}$$

Then, the threshold Th is calculated by

$$Th = argmax(\mathrm{e}) = \frac{(\Delta\mu(s)/\mu(s)) + (\Delta\sigma(s)/\sigma(s))}{|\mathrm{s}_a| + |\mathrm{E}_{sea}^2|} \tag{14}$$

where $\Delta\mu(s) = \mu(s) - \mu(s < Th)$ and $\Delta\sigma(s) = \sigma(s) - \sigma(s < Th)$, NDT also considers preventing generating a large number of anomalous values $\mathbf{s}_a = \{s > Th\}$ and anomalous sequences $|\mathrm{E}_{sea}^2|$. NDT aims to identify a threshold that results in the most significant decrease in the average and variability of $\mathbf{s}$ by eliminating any values over the threshold.

4 Evaluation

4.1 Experiment Setting

4.1.1 Dataset. We conducted experimental research on both open-source datasets and datasets generated from real-world systems.

Open-source Microservice System: We utilize three open-source microservice benchmarks, namely AIOPS 2022 [4], Generic

Table 3: The effectiveness of FUSION in comparison with different methods on three datasets.

Approach	AIOps 2022			GAIA			SN			MMS		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
TraceVAE	0.423	0.406	0.414	0.411	0.396	0.403	0.532	0.511	0.521	0.375	0.321	0.346
DeepTraLog	0.689	0.652	0.670	0.715	0.694	0.704	0.721	0.706	0.713	0.521	0.503	0.512
Hades	0.712	0.694	0.703	0.728	0.719	0.723	0.733	0.724	0.728	0.534	0.521	0.527
MADMM	0.727	0.731	0.729	0.749	0.758	0.753	0.754	0.763	0.758	0.563	0.544	0.553
MISP	0.758	0.736	0.747	0.772	0.761	0.766	0.786	0.767	0.776	0.594	0.547	0.570
AnoFusion	0.783	0.769	0.776	0.794	0.785	0.789	0.784	0.752	0.768	0.694	0.708	0.701
Medicine	0.808	0.793	0.800	0.814	0.797	0.805	0.817	0.804	0.810	0.727	0.711	0.719
Triple	0.843	0.829	0.836	0.854	0.832	0.843	0.918	0.907	0.912	0.827	0.815	0.821
MSTGAD	0.932	0.918	0.925	0.921	0.927	0.924	0.952	0.938	0.945	0.857	0.831	0.844
FUSION w/o Wavelet	0.842	0.834	0.838	0.863	0.873	0.868	0.896	0.872	0.884	0.778	0.763	0.770
FUSION w/o Temporal	0.724	0.713	0.718	0.734	0.727	0.730	0.718	0.704	0.711	0.657	0.621	0.638
FUSION w/o Spatial	0.793	0.779	0.786	0.784	0.768	0.776	0.828	0.813	0.820	0.727	0.735	0.731
FUSION w/o U-structure	0.648	0.618	0.633	0.692	0.682	0.687	0.739	0.716	0.727	0.618	0.603	0.610
FUSION	**0.959**	**0.941**	**0.950**	**0.963**	**0.952**	**0.957**	**0.978**	**0.963**	**0.970**	**0.939**	**0.921**	**0.930**

Table 4: Comparison of efficiency between FUSION and different methods. Training time (Offline(min)) and average time to diagnose a failure case (Online(s)).

Dataset		TraceVAE	DeepTraLog	Hades	MADMM	MISP	Medicine	AnoFusion	Triple	MSTGAD	**FUSION**
AIOps 2022	Offline	19.25	77.28	39.00	34.45	112.46	43.91	106.96	47.80	68.32	**41.23**
	Online	8.41	148.15	10.52	14.38	193.71	15.11	184.91	18.96	21.01	**14.22**
GAIA	Offline	17.42	68.26	28.91	24.63	109.00	31.45	97.06	32.61	54.98	**29.75**
	Online	6.51	125.71	5.68	7.03	164.41	10.45	157.60	14.64	17.30	**11.61**
SN	Offline	5.06	56.57	8.30	14.16	87.49	17.73	82.25	19.83	46.58	**20.68**
	Online	1.21	118.95	2.38	4.89	134.88	6.59	125.10	8.10	10.23	**6.17**
MMS	Offline	26.23	89.36	49.47	58.90	148.97	64.12	137.01	69.63	78.67	**65.45**
	Online	13.45	167.77	37.16	43.05	232.92	59.82	221.56	67.77	85.57	**47.22**

AIOps Atlas (GAIA) [5], and SocialNetwork [9] (SN,) which has been widely used in the existing work [3, 14, 17, 18, 29]. AIOPS 2022 is constructed by China Construction Bank based on hiptershop system. GAIA is collected from a system utilized by both mobile and PC users. SN is derived from an open-source microservice system that implements a broadcast-style social networking platform.

Microservices-based Mail System: This system is a real-world mail system (MMS) with 56 microservices in a large Internet company with more than two million users (part of its whole system). The typical faults include the following: CPU stress, memory stress, high disk I/O latency, disk partition full, database failure, resource exhaustion, high packet loss rate, etc.

4.1.2 Evaluation Metrics & Baselines. To evaluate the accuracy of anomaly detection, following the common practice in anomaly detection [19, 22, 26, 32], we choose Precision, Recall, and F1 to evaluate the effectiveness of anomaly methods based on TP (True Positive), FP (False Positive), and FN (False Negative).

To comprehensively evaluate the anomaly detection performance of the model, we selected nine representative baselines for comparison: TraceVAE [28], DeepTraLog [30], Hades [15], MADMM [26], MISP [19], Medicine [25], AnoFusion [32], Triple [22] and MSTGAD [11]. It is important to note that the original method is supervised in the Medicine [25] baseline. We retained the steps for multimodal processing and feature extraction in the original method and replaced the final fully connected layer with an unsupervised clustering approach, the same as DeepTraLog [30].

Table 5: Performance Comparison of FUSION and Baseline Methods under Noise $\sigma = 0.1$ Across Different Modalities.

Approach	Metric_noise		Log_noise		Trace_noise	
	F1	Decrease	F1	Decrease	F1	Decrease
TraceVAE		×		×	0.331	-4.31%
DeepTraLog		×	0.469	-8.37%	0.434	-15.21%
Hades	0.468	-11.27%	0.471	-10.70%		×
MADMM	0.477	-13.80%	0.492	-11.08%		×
MISP	0.493	-13.44%	0.502	-11.86%		×
AnoFusion	0.632	-9.83%	0.634	-9.55%	0.588	-16.11%
Medicine	0.672	-6.53%	0.671	-6.66%	0.639	-11.12%
Triple	0.747	-9.01%	0.623	-24.11%	0.734	-10.59%
MSTGAD	0.809	-4.12%	0.722	-14.43%	0.774	-8.27%
FUSION	**0.913**	**-1.82%**	**0.879**	**-5.48%**	**0.908**	**-2.36%**

4.1.3 Implementation Details. As for hyperparameters, we set the batch size as 32 and use the Adam optimizer. In our multi-resolution analysis of multimodal data using DWT, we chose the Haar [16]

Table 6: Performance Comparison of FUSION and Baseline Methods under Noise $\sigma = 0.2$ Across Different Modalities.

Approach	Metric_noise		Log_noise		Trace_noise	
	F1	Decrease	F1	Decrease	F1	Decrease
TraceVAE		×		×	0.328	-5.18%
DeepTraLog		×	0.458	-10.52%	0.412	-19.51%
Hades	0.448	-15.06%	0.462	-12.40%		×
MADMM	0.439	-20.66%	0.483	-12.71%		×
MISP	0.423	-25.73%	0.479	-15.90%		×
AnoFusion	0.583	-16.82%	0.605	-13.69%	0.558	-20.39%
Medicine	0.613	-14.73%	0.645	-10.28%	0.613	-14.73%
Triple	0.689	-16.07%	0.553	-32.64%	0.684	-16.68%
MSTGAD	0.751	-11.00%	0.698	-17.28%	0.745	-11.71%
FUSION	**0.892**	**-4.08%**	**0.864**	**-7.09%**	**0.889**	**-4.40%**

wavelet as the basis function and conducted a three-level decomposition of the time series data, employing high-pass and low-pass filters to produce approximation and detail coefficients at different resolutions. As for the diffusion network, we adopt the following quadratic schedule for variance schedule $\beta_n = \left(\frac{N-n}{N-1}\sqrt{\beta_1} + \frac{n-1}{N-1}\sqrt{\beta_N}\right)^2$. We set the minimum noise level $\beta_1 = 0.0001$ and hidden size $C = 96$ and search the maximum number of the diffusion step N and the maximum noise level β_N from a given parameter space ($N \in [50, 100, 200]$, and $\beta_N \in [0, 1, 0.2, 0.3, 0.4]$), and each model's best performance is reported in the experiment. The history length T_h and prediction length T_p are set to 100 and 80 for all datasets.

4.2 Performance Comparison

We conducted experiments comparing multiple baseline methods across four datasets. As shown in Table 3, FUSION outperformed all probabilistic baselines, improving the F1-score by 2.70%, 3.62%, 2.70%, and 10.21% on the respective datasets. Bimodal methods usually underperform due to their limited analysis of multimodal data. All approaches performed best on the SN dataset, which is simpler and lower in dimensionality, while the PMS dataset showed the worst results due to its complexity and noise. MSTGAD ranked second on most datasets due to its semi-supervised learning and attention mechanisms, but FUSION consistently achieved the best performance by effectively extracting multimodal features and incorporating uncertainty through diffusion models.

Inference Time. We compared the training and detection times of baseline models in Table 3. TraceVAE, Hades, and MADMM are faster due to processing fewer modalities, while MISP and AnoFusion analyze multimodal data, resulting in longer times. DeepTraLog's approach of training on individual traces before anomaly detection also increases time consumption. Diffusion-based models like TimeGrad [21] and DDPM [10] are slower because of their iterative denoising process but show promising performance. Our proposed FUSION accelerates by using a non-autoregressive architecture and randomized diffusion steps. Increasing α improves efficiency without compromising performance when P_s is large. While FUSION may not significantly speed up compared to the models in Table 3, it meets the requirements for online diagnostics, showcasing its potential for practical applications.

Noise Analysis. FUSION demonstrates the best performance when dealing with Gaussian noise across various modalities, highlighting its robustness and effectiveness in Table 5 and 6. When noise is added to trace data, TraceVAE ranks second best due to its ability to capture latent structures and uncertainty, providing noise robustness. In contrast, when noise affects metric data, MSTGAD performs second best, leveraging its spatio-temporal attention and semi-supervised learning. Additionally, Medicine mitigates log noise by assigning different attention weights to modalities. FUSION combines multimodal data in the wavelet domain, integrating features while preserving unique information from each modality. Even if one modality experiences noise or interference, information from other modalities can still support accurate anomaly detection

4.3 Ablation Study

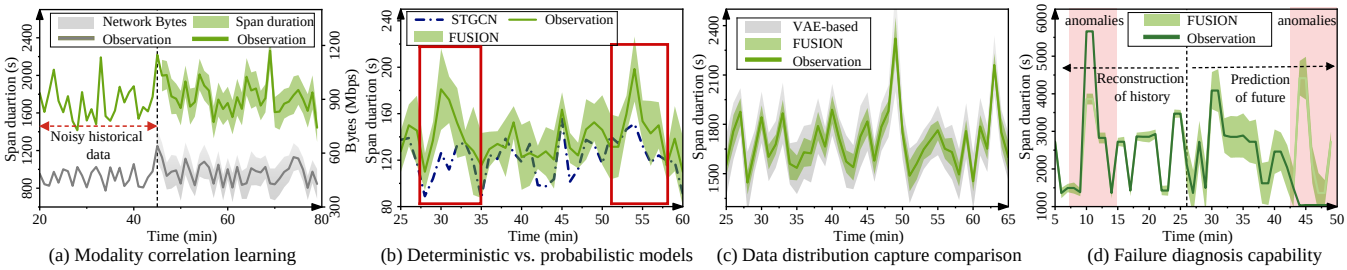
We perform an ablation study using four datasets to assess the impact of each component. We have developed four variants of FUSION and compared them to the full version of FUSION, as shown in Table 3. The distinctions are detailed below:

- **w/o Wavelet:** Remove the wavelet fusion. The multimodal serialization is directly sent to the diffusion network.
- **w/o Temporal:** Remove the TCN component from the U-structure without modeling the temporal correlation.
- **w/o Spatial:** Remove the GNN component from the U-structure without modeling the spatial correlation.
- **w/o U-structure:** Remove the Unet-based structure and use only one TCN for feature extraction.

The U-structure greatly influences model performance, with its removal leading to a notable decline. This highlights the U-structure's strength in capturing spatiotemporal dependencies. Additionally, neglecting either spatial or temporal dependency learning significantly impacts performance, reinforcing the need to model spatiotemporal correlations. The removal of wavelet-domain feature fusion has the most significant effect on the MMS dataset, as real-world modal data is susceptible to noise or interference. However, wavelet domain multimodal data fusion effectively integrates information, allowing the model to maintain accuracy in anomaly detection even when one modality is compromised.

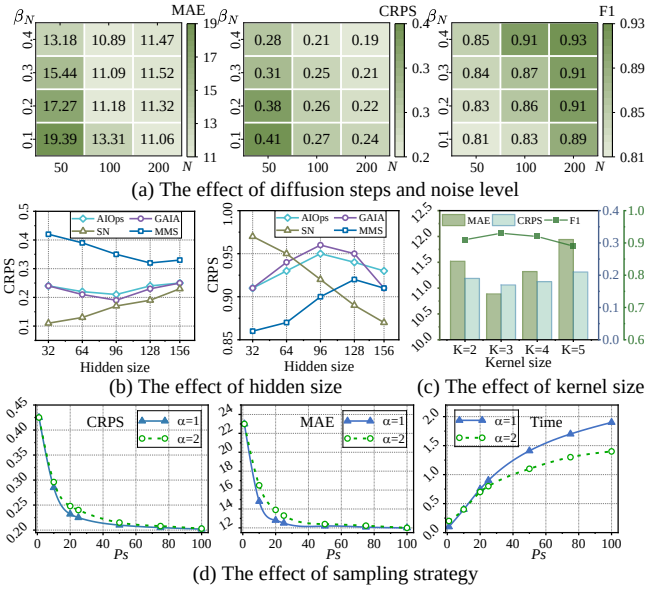
4.4 Interpretability of Failure Diagnosis

We illustrate the performance of FUSION using the MMS dataset: **(1) Solutions to limitation 1.** We validate FUSION's ability to learn from multiple modalities by visualizing network byte distributions and span durations in Figure 3(a). Even with noise in the T_h time series, the model retains unique information from each modality through frequency-domain feature fusion, ensuring that if one modality fails, the others still support anomaly detection. **(2) Solutions to limitation 2.** We compared FUSION with MSTGAD using the STGCN framework. FUSION offers a tighter prediction interval, shown in Figure 3(b), indicating greater reliability in capturing average behavior and data variability. Additionally, it more accurately represents data distribution than VAE-based models like TraceVAE in Figure 3(c), thanks to its diffusion model's effective step-by-step denoising process that captures complex multimodal patterns. **(3)**



(a) Modality correlation learning (b) Deterministic vs. probabilistic models (c) Data distribution capture comparison (d) Failure diagnosis capability

Figure 3: Successful cases of FUSION in forecasting and anomaly evaluation.



(a) The effect of diffusion steps and noise level

(b) The effect of hidden size (c) The effect of kernel size

(d) The effect of sampling strategy

Figure 4: The effect of hyperparameters.

Failure diagnosis capability. The model in Figure 3(d) reconstructs historical data and predicts future states. This dual functionality facilitates tracing fault causes, provides early warnings, and supports informed preventive maintenance, thereby enhancing system reliability and operational efficiency.

4.5 Hyperparameters Sensitivity

Effect of the diffusion network hyperparameters. We explore how the maximum number of diffusion steps N and the maximum noise level β_N from the parameter space ($N \in \{50, 100, 200\}, \beta_N \in \{0, 0.2, 0.3, 0.4\}$) impact our results. Increasing diffusion steps enhances generation quality but also raises computational costs and training time. Reducing steps speeds up processes but may compromise quality. Larger noise increases robustness and diversity but introduces uncertainty, while smaller noise improves stability and quality but limits exploration. Figure 4(a) shows that with $N = 50$ diffusion steps and $\beta_N = 0.1$, model performance drops significantly due to insufficient steps, leading to a poor approximation of the Gaussian distribution and degraded performance.

Effect of hidden size and kernel size. Experiments show that model performance improves with hidden layer size up to a point, after which it declines. Optimal sizes vary by dataset: 32 for SN, 96 for AIOPS and GAIA, and 128 for MMS (Figure 4(b)). The MMS dataset requires a larger hidden layer due to its complexity. In TCN, the convolution kernel size ($K = 2, 3, 4, 5$) affects time step coverage (Figure 4(c)). Larger kernels capture longer dependencies but increase computation. For $N = 200$ and $\beta_N = 0.4$, a kernel size of $K = 3$ is optimal for the MMS dataset.

Effect of hidden size and kernel size. The parameter α is suggested to improve both diversity and performance in model training. As shown in Figure 4(d), increasing sample size P_s enhances accuracy, particularly between sizes 20 and 30, with diminishing returns beyond 40. In reverse diffusion, α affects diversity and efficiency. A smaller α promotes greater diversity and data exploration, while a larger α increases efficiency by processing fewer samples, which is useful for quick predictions. Figure 4(d) compares $\alpha = 1$ and $\alpha = 2$. For larger sample sizes ($P_s > 40$), $\alpha = 2$ matches $\alpha = 1$ in performance but is 1.5 times faster. For smaller sample sizes, a smaller α is advisable to enhance both diversity and performance.

5 Conclusion

This study underscores the critical role of multimodal data analysis in diagnosing anomalies within microservices. We decompose and fuse multimodal data in the wavelet domain. We've developed an attention mechanism that enhances key features based on Pearson correlation coefficients. These coefficients are then processed and reconstructed into enhanced feature representations using IDWT. We propose a spatiotemporal graph-based Denoising Diffusion Probabilistic Model that integrates spatiotemporal learning with DDPM's uncertainty quantification to capture spatiotemporal features. FUSION, utilizing a U-structure architecture, captures multi-scale temporal dependencies and models spatial correlations with Graph Neural Networks. FUSION's non-autoregressive approach efficiently generates multi-horizon predictions, enhancing fault diagnosis in microservices. Our experiments show that this approach effectively deals with complex data patterns and noise found in real-world microservice environments. FUSION offers a strong and efficient solution for detecting and diagnosing anomalies by combining different data analysis methods and advanced modeling, leading to more reliable and scalable microservice operations.

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